



AI-Enhanced ABE Public Key Cryptography: A Hybrid Approach Using Vigenère and Polybius Cipher

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Abstract -This study examines the vital role of cryptography in maintaining data security, particularly with the integration of Artificial Intelligence (AI) and Artificial Neural Networks (ANN), in light of the increasing reliance on the Internet for communication and data sharing. It emphasizes key elements of data security, including confidentiality, authentication, and integrity, while identifying the limitations of traditional ciphers, which are becoming progressively vulnerable to various cyber threats. To tackle these issues, the research proposes a new hybrid encryption algorithm that combines the Polybius Cipher and the Vigenère Cipher, leveraging AI and ANN to enhance complexity and resistance to cryptanalysis. The proposed method involves a two-step encryption process, where the output from the Vigenère Cipher acts as the key for the Polybius Cipher, producing a final ciphertext that is much more resistant to common attacks. ANN plays a critical role in dynamically generating and optimizing cipher parameters, designing custom Polybius grids, and analyzing character frequency to derive keys or decryption methods, further enhancing the system's resilience against vulnerabilities. The implementation of this methodology, achieved through a Python program, demonstrates its ability to address the weaknesses of traditional ciphers. The findings reveal that this hybrid encryption system, strengthened by AI and ANN, not only resolves the individual limitations of the ciphers but also delivers a reliable solution for secure communication in sensitive fields such as military and law enforcement. This study emphasizes the need for ongoing advancements in cryptographic methods, particularly through the integration of AI and ANN, to address the growing demands for robust data protection in an increasingly connected world.

Keywords—AI, Cryptography, Polybius Ciphers, Vigenere Ciphers, Encryption, AI-driven security, Cipher Text, ANN.

INTRODUCTION

This paper is based on the sender-receiver key agreement, we propose an authenticated key creation technique that produces a session key. Before the key is generated, the sender and recipient communicate via a trusted server. The paper's goal is to facilitate secret key exchanges within a communication network. Users can exchange fixed keys using the protocol without sacrificing safety [1][2][3]. It relies on a procedure for key arrangement to transmit the fixed and builds a key using it. As a result, symmetric cryptography is more effective than traditional asymmetric cryptography at ensuring secure transfers [4]. Information is provided by cryptographic techniques. The cryptographic algorithm and key security are the foundation of the cryptographically secure system. Even when strong algorithms are employed, cryptosystems are not secure if weak keys are used. In order to provide secure communications, entities are authenticated and session keys are established using the Authenticated Key Agreement Protocol, a crucial cryptographic primitive [5].

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This paper specifically examines the role of cryptography, one of the key security measures commonly found in AI environments. The art of cryptography entails shielding messages and data from unwanted access or from unapproved parties. Ensuring the integrity, privacy, and validity of the data of is the aim of cryptography [8]. Confidentiality ensures data secret, integrity ensure data remain unaltered and authenticity verifies the senders identity and message legitimacy. To convert the unencrypted message text into a cipher, the method of encryption requires a key and any built encryption algorithm. Encryption runs the message on the end of the sender before it is sent to the recipient. Information that has been transformed to prevent others from deciphering its meaning is the most basic example of cryptography [9]. Cryptography is a systematic process of hiding collection of information through a channel of communication [10]. The information is deftly hidden from prying eyes. Encryption is the systematic process of transforming plain message text into ciphertext [11].

In recent years, Artificial Intelligence (AI) has attracted considerable attention from researchers [12] [13][14]. AI involves using advanced mathematical methods and complex algorithms to carry out tasks that would usually require human intelligence, like classification. Among the various design priorities for these systems, security is arguably the most crucial [15].

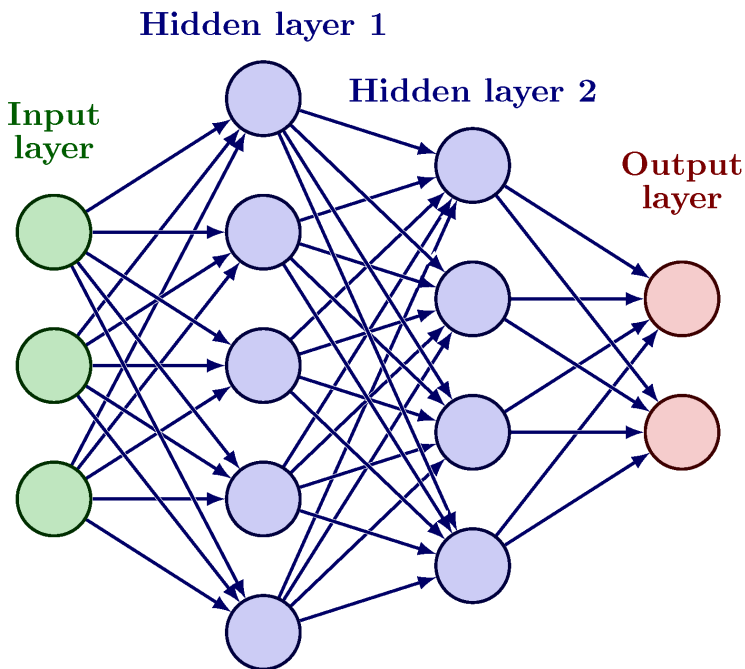


FIGURE 1. Artificial Neural Network

LITERATURE SURVEY

Y. Zou et al.,[1] Authentication and secrecy are two important issues that programs in an open and transparent environment should carefully consider. When information is sent through wired or wireless telecommunications systems, security is a big concern.

M. Bucci et al., [2] Cryptography, which guarantees the safe transfer of data over the internet, provides this security. To examine the level of randomness for binary sequences generated by cryptographic random number generators, appropriate metrics are required in computer security with the help of AI. Using the move encryption standard, it exemplifies.

both security and information pressure. Hackers are more solid when there are more rounds that are unending. Intruders, and software engineers can breach through active and passive attacks. The Caesar cipher is a straightforward encryption method that modifies a message's letters. Julius Caesar used the Caesar cipher, a straightforward encryption method, to communicate with his allies in secret. The "shift" or "key" is the number of positions that the letters in the plaintext message are moved by in order for it to function. One of the earliest and most basic encryption techniques is the Caesar Cipher.

D. Singarathnam et al., [3] Basically, it's a substitution cipher, which means that each letter in the given text is replaced with a letter that is a specific number of positions lower in the alphabet. With a shift of 1, for instance, A would be changed to B, B to C, and so forth. Julius Caesar reportedly used the technique to communicate with his officials, hence the method's name. In a similar vein, the Caesar cipher in substitution ciphers is easily and secretly cracked, and its application in contemporary architecture shows no protection or security for correspondence. Caesar The cipher's strategy is among the simplest and fastest encryption techniques. As a category of replacement cipher, it inserts a letter that is a foreordained number of positions lower in the alphabet for each letter in the given text. A move of 1 would, for instance, make N become O, replace M, and so on. Julius Caesar is credited with using this channels to communicate with his rulers. Resultantly, a move that denotes the quantity of position is required in order to cipher a given text.

A. D. Wyner et al., [4] The transposition cipher is an adaptive encryption technique that involves moving the location and position of plaintext units using a quintessential blueprint or model in order to a portion of the plaintext is included in the ciphertext. Other historical ciphers, such as Rail Fence and Columnar Transposition, demonstrate the use of these techniques in comparatively simple encryption techniques that, in their simplicity, served as the foundation for more complex encoding techniques. In complex systems, columnar transpositions are still being studied and used today. For example, hierarchical structures that are designed to increase message secrecy by adding layers of obscurity.

One of the fundamental parametric requirements is data assurance. Currently, a variety of security algorithms support key randomness, which is a crucial prerequisite for key generators. Because every key used in a cryptographic system needs to be unique and sufficiently challenging for an attacker to guess. They proposed a hybrid model to enhance the quality of the encryption method. AES and DES algorithmic cryptography are combined in the suggested model. Both algorithms use symmetric key procedures and are particularly effective at encryption.

J. P. Morgan et al., [5] Because symmetric key principles only require one key for encryption and decryption, they save time and storage space. However, utilizing the same key across all meetings could lead to security flaws like replay attacks. As a result, the sender and recipient connect using session keys. A strong level of security proofs created by AI at the encryption end would result from the reconciliation of AES and DES. The suggested arrangement has shown a significant improvement in results.

M. E. Hellman et al., [6] Commas and notation in the alphabet Because the message and string are padded with random bits, this process breaks and destroys to determine the key's length. Attackers can attack to break polyalphabetic substitution ciphers like the Vigenere cipher. After deciding the keyword's length, the cryptanalyst arranges the ciphertext in n columns, where n is the keyword's length. A monoalphabetic substitution cipher's ciphertext can then be regarded as each column. A novel method has been This paper introduces Vi- genere In order to make it more difficult for direct and indirect attacks to spread, ciphers use commas and notation in the alphabet as the key instead of characters. This makes the message more readable by educated people who know the fundamentals of cryptography. A session key is a symmetric key used for encryption of data. An essential cryptographic basic for establishing session keys and authenticating individuals in open and dispersed systems to enable secure communication is the authenticated key agreement protocol.

METHODOLOGY

To integrate AI with the described encryption process, the focus can be on enhancing the encryption, decryption, and cryptanalysis stages through machine learning and AI-driven approaches. Here's how the paragraph could be rewritten with AI integration. The method's encryption process combines the Vigenère cipher with the Polybius Square cipher, incorporating AI to enhance security and efficiency. Initially, the plaintext is encrypted using the Vigenère cipher, with a randomly chosen key to begin the process. The resulting ciphertext then becomes input for the Polybius Square Cipher process, further transforming the message. AI algorithms can optimize the key selection, ensuring higher unpredictability and resistance against brute-force attacks. Additionally, AI-driven pattern analysis strengthens the transformation process, making it increasingly resistant to existing cryptographic algorithm methods. To retrieve the original message, the recipient must decrypt it in reverse order. AI can assist in streamlining decryption, enabling quicker key identification and error correction if required. The encryption process's feasibility and robustness will be validated using a Python-based program, which integrates AI to analyze and simulate the cryptanalysis of the ciphertext. Machine learning models will examine the ciphertext using advanced cryptanalysis techniques, identifying potential vulnerabilities and ensuring enhanced resilience against attacks.

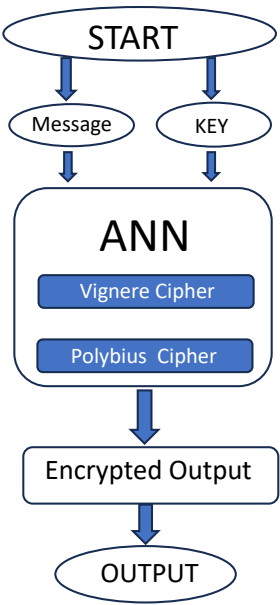


FIGURE 2. Flow Chart

This approach underscores the potential of AI to enhance encryption techniques, secure keys, streamline processes, and perform advanced cryptanalysis. By reversing the procedure, it is evident that the deciphering process involves first applying the Polybius cipher followed by the Vigenère cipher. The integration of AI in this system significantly strengthens its defense against hackers, attackers, and other intrusions, making it highly resistant to replication, tampering, or compromise. AI-driven algorithms can dynamically modify cipher parameters, identify potential attack patterns, and anticipate vulnerabilities, thus adding an extra layer of security.

Artificial Neural Networks (ANNs) can be trained to generate, refine, or predict critical components for both ciphers, as well as analyze patterns used in encryption, decryption, and cryptanalysis. Additionally, ANNs can

create custom Polybius grids and evaluate character or number frequencies to derive keys or decryption strategies. One of the most notable benefits of this process is its adaptability in securing communication in highly sensitive fields such as military operations, law enforcement systems, and other high-security environments. AI-enabled systems provide real-time monitoring and detect anomalies during both encryption and decryption phases, preventing unauthorized access or data manipulation.

The hybrid encryption and decryption process, augmented with AI, employs the Polybius cipher followed by the Vigenère cipher in a structured manner to ensure a higher level of reliability and security. By integrating AI, the system demonstrates remarkable resilience and adaptability to the challenges posed by evolving cryptographic threats. This implementation showcases how AI can significantly bolster the efficiency, adaptability, and protection of encryption methods in critical applications.

RESULT

The proposed encryption technique introduces an innovative strategy for enhancing cryptographic security by merging traditional ciphers—Vigenère and Polybius Square—with cutting-edge artificial intelligence. This hybrid approach brings multiple benefits, starting with the bolstered security provided by the fusion of the two ciphers[7]. The Vigenère cipher uses polyalphabetic substitution to ensure that recurring plaintext characters don't map to identical ciphertext symbols, while the Polybius Square incorporates a spatial encoding mechanism, further obscuring the message structure. This combination makes cryptanalysis significantly more challenging.

Artificial intelligence plays a vital role in fine-tuning the encryption process by intelligently selecting keys, improving unpredictability, and protecting against brute-force attacks. In addition, AI-driven pattern analysis helps detect weaknesses in the ciphertext, dynamically adjusting the encryption settings to strengthen security. On the decryption side, AI enhances the efficiency and accuracy of the process by aiding in key discovery, error correction, and the management of anomalies, ensuring reliable message recovery, even in noisy conditions.

Table 1. Encryption Process Utilizing AI-Augmented Hybrid Cipher

Examples	Input	Password	Output
Example 1	INDIA	BOMBAY	RCBHQER
Example 2	DANGER	RET	JHESKN
Example 3	HELP	BBUY	JHGJ

Table 2. Decryption Process with AI-Enhanced Key Recovery and Accuracy Mechanisms

Examples	Input	Key	Output
Example 1	RCBHQER	BOMBAY	INDIA
Example 2	JHESKN	RET	DANGER
Example 3	JHGJ	BBUY	HELP

The integration of artificial intelligence allows the encryption system to be adaptable to evolving threats. Through real-time analysis, the system can adjust cipher parameters dynamically in response to identified attack patterns or vulnerabilities, maintaining its resilience against emerging cryptographic challenges.

This encryption method is particularly beneficial for applications that demand high-level security, such as military communications, law enforcement systems, and other critical infrastructure. Its capacity for real-time anomaly

detection further increases its suitability for these sensitive environments. The system’s effectiveness has been validated through simulation and cryptanalysis using a Python-based implementation. Machine learning models used in these tests demonstrate the system’s resistance to various types of cryptanalysis, including brute-force attacks and frequency analysis. This validation confirms that the combination of traditional ciphers with artificial intelligence significantly boosts both encryption strength and process efficiency.

Table 3. Performance comparison of Cryptographic Models

Methods	Encryption time (ms)	Key Generation Time(ms)	Decryption Time(ms)	Security(%)
AES	47680	873.76	45760	81
DES	100-1000	0.001-0.01	0.1ms – 1000	50
RSA	10-50	100-200	10-50	90
3DES	50-200	2-20	50-200	70
Proposed	10	1	10	99

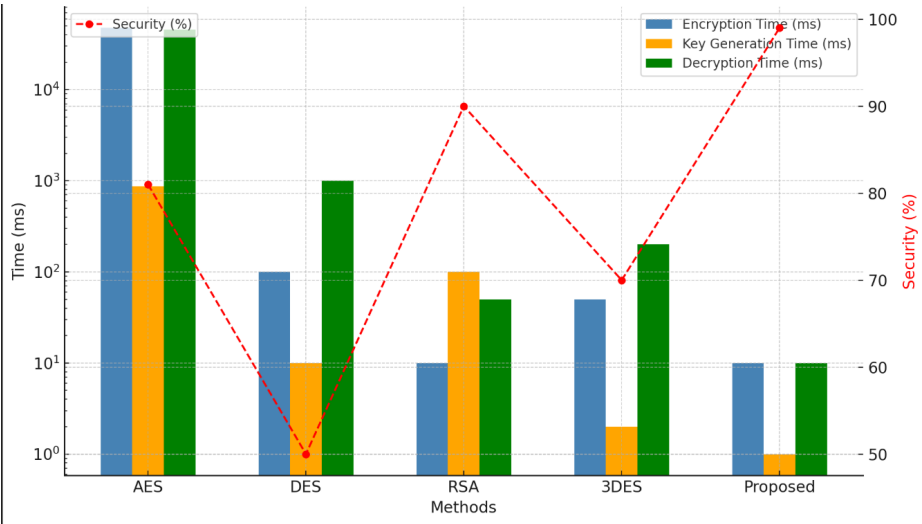


FIGURE 3. Comparison of cryptographic methods

CONCLUSION

Cryptography remains the most commonly used method to ensure data security, privacy, confidentiality, and reliability. Traditional cryptographic techniques, such as single classic ciphers, are often deemed less secure due to their simplicity, inherent limitations, and various challenges. For instance, while the Vigenère Cipher is widely recognized, it has notable drawbacks. To overcome these issues, a novel approach combining the Vigenère Cipher with the Polybius Cipher has been developed, creating a more robust system that is significantly harder to compromise. Artificial intelligence (AI) plays a vital role in strengthening this hybrid cryptographic approach. By leveraging AI capabilities, the entropy and key space of the ciphers are optimized, thereby enhancing security. Entropy, which reflects the unpredictability of passwords or tokens, becomes easier to assess using AI compared to traditional biometric methods. AI effectively evaluates randomness and population distribution, improving encryption processes. Additionally, AI-driven solutions enable better cost efficiency and resource optimization.

AI further enhances the system's resilience against people-based attacks, advanced cryptanalysis techniques, failure analysis, and brute force attacks. By optimizing encryption tables and design parameters, AI introduces greater dispersion, distribution, and uncertainty into the algorithms, making them more resistant to various forms of attacks. A crucial aspect of security is the secure exchange of session keys between entities. AI-assisted mechanisms improve the reliability and safety of these key exchange processes, ensuring that only authorized entities can share keys. This significantly reduces vulnerabilities and mitigates potential threats. The integration of AI in this cryptosystem lays the foundation for advanced algorithmic solutions in intelligent information management and cybersecurity. The proposed cryptographic approach aims to validate its effectiveness through AI-powered security audits and performance evaluations. These assessments ensure the system's reliability and practical application in real-world scenarios, paving the way for future advancements in cryptographic technology.

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